Protein-Protein Interaction Network Analysis for Endometriosis

Social Network Analysis course final project

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Research questions

- Which set of k proteins, blocked together, best limits the spread of the disease signal?
- Does the optimal set depend on the chosen algorithm?

Methodology

Diffusion models:

- Susceptible-Infected-Recovered (SIR),
- Independent Cascade (IC).

Blocking algorithms:

- group centrality: greedy group degree, greedy iterative betweenness,
- influence maximization: greedy, run under SIR and IC

Data Sources and Network Construction

Endometriosis Genes from DisGeNET: https://disgenet.com

- Filters: UMLS C0014175, confidence score ≥ 0.25, 166 genes total

Protein graph from STRING database: https://string-db.org

- Filters: confidence score ≥ 0.4, first-shell expansion, dropped weights

Network Results

- unweighted,
- undirected,
- connected,
- 454 nodes,
- 21,095 edges,
- average degree ≈ 93,
- density ≈ 0.205

Null models

Random networks as a control group, all share $V = 454$.

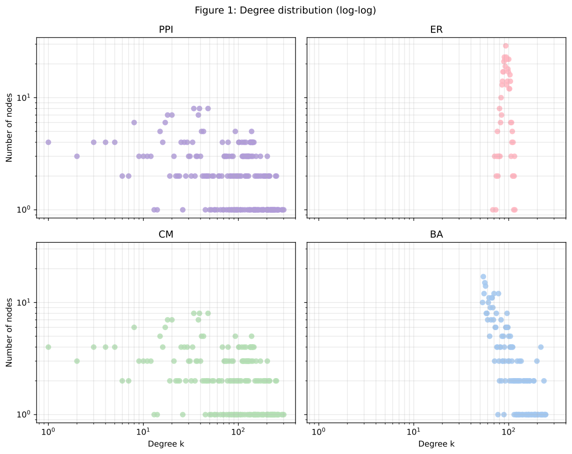
- Erdős-Rényi
- Barabási-Albert
- Configuration model

Network statistics

	$\lvert E \rvert$	Density	Clustering	$\bar{\ell}$
PPI	21,095	0.205	0.637	1.94
ER	21,253	0.207	0.206	1.79
BA	21,253	0.207	0.298	1.79
CM	21,095	0.205	0.512	1.84

PPI clustering stands out

Degree distribution



PPI and CM heavy-tailed; ER narrow Poisson; BA power-law;

Diffusion models

Simulate the disease signal spreading and score blocking strategies by how much they reduce propagation.

$$\text{Spread reduction} = 1 - \frac{\text{spread for the blocked network}}{\text{spread for the unblocked network}}$$

- SIR, Susceptible-Infected-Recovered: infection rate β, recovery rate γ.
- IC, Independent Cascade: activation probability p.

Diffusion models' calibration

Calibrate β, γ, p so that unblocked propagation reaches around 70% of the network.

Parameters	Mean spread	% of network
SIR $\beta = 0.005, \gamma = 0.20$	318.7	70
SIR $\beta = 0.010, \gamma = 0.20$	382.3	84
SIR $\beta = 0.020, \gamma = 0.20$	418.6	92
SIR $\beta = 0.050, \gamma = 0.20$	439.5	97
IC $p = 0.005$	59.9	13
IC $p = 0.010$	170.5	38
IC $p = 0.020$	294.7	65
IC $p = 0.050$	386.0	85

Chosen: $\beta = 0.005$, $\gamma = 0.20$, $p = 0.02$.

Blocking algorithms

Three algorithms choose the proteins to block: two group-centrality and one influence-maximization.

- `gdeg`, greedy group degree: removing hubs,
- `gbtw`, greedy iterative betweenness: removing bridges,
- `gim`, greedy influence maximization: reducing spread

`gim` runs twice, once under SIR and once under IC.

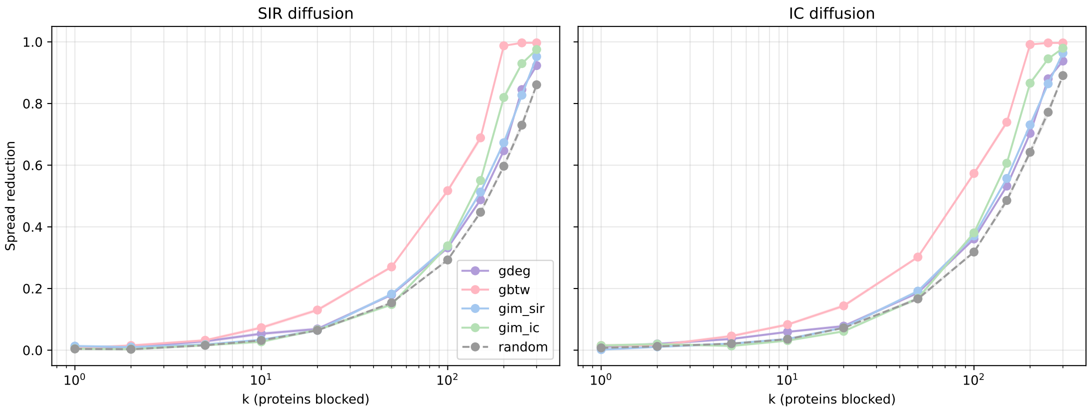
Top 10 picks per chain

	gdeg	gbtw	gim_sir	gim_ic
1	IL6	ALB	FGF4	CD40
2	ESR1	ESR1	KDR	LAG3
3	AKR1C3	EGFR	UGT1A7	CX3CL1
4	NCOA1	IL6	VTN	HDAC2
5	EGFR	GAPDH	IL17A	MED4
6	INS	TNF	NTRK3	UGT2B11
7	ITGB3	IL1B	PDGFRB	MED14
8	APOE	INS	SERPINF1	CCL13
9	CTNNB1	AKT1	EREG	PTPRC
10	CYP3A4	ACTB	TIMP1	COPS2

ESR1: estrogen receptor; FGF4, KDR: angiogenesis; CD40: immune.

Spread reduction across strategies

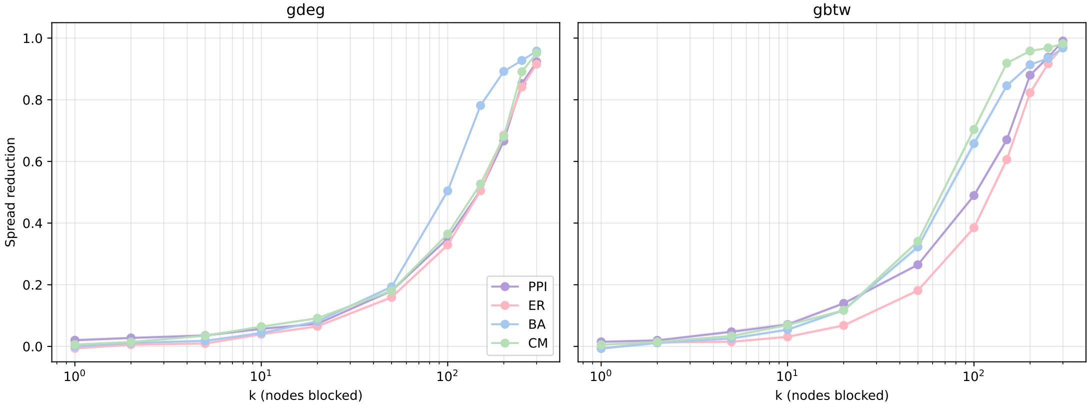
Figure 2: Spread reduction vs blocked-set size, shaded band shows SEM



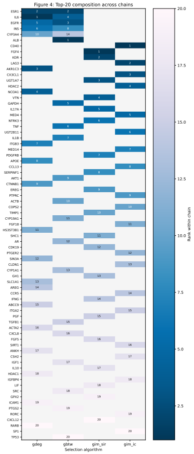
`gbtw` dominates; saturates at $k \approx 200$ with 99% reduction.

PPI vs null networks

Figure 3: Spread reduction on PPI vs null networks, shaded band shows SEM



Top-20 composition



75 distinct proteins; only 5 shared by 2+ chains; 93% algorithm-specific.

Conclusion

The answer to the research question:

- Best k-set: prefix of the `gbtw` chain, saturates near $k = 200$
- Strongly algorithm-dependent: 93% of top-20 picks are algorithm-specific

At a realistic small k, partial blocking reduction is not significant at all.